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Decision Making with Business Statistics

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1. Introduction

Decision-making in real-world business and operational environments is rarely straightforward. Managers and practitioners are often required to make choices under uncertainty, where outcomes are influenced by multiple interacting factors rather than a single cause. In such situations, relying only on intuition or experience can lead to inconsistent results. Business statistics provides a structured approach to understanding data, identifying meaningful patterns, and supporting decisions with evidence.

This report applies statistical techniques to four independent problem statements drawn from practical contexts such as sports management, manufacturing quality control, material suitability assessment, and healthcare operations. Each problem represents a common decision-making challenge where variability, risk, and uncertainty play a key role. The goal of the analysis is not only to compute statistical measures but to interpret them in a way that supports clear and actionable business decisions.

Throughout the report, tools such as probability analysis, hypothesis testing, and analysis of variance (ANOVA) are used to examine relationships between variables and assess whether observed differences are statistically meaningful. Visual representations generated using Python are included alongside numerical results to improve clarity and help communicate insights more effectively. These visuals play an important role in highlighting trends, differences, and interaction effects that may not be immediately obvious from tables alone.

Rather than treating statistics as a purely theoretical exercise, this report emphasizes practical interpretation. Each section links statistical outcomes back to the underlying business equation, helping translate results into recommendations that could realistically guide policy, quality control, or process improvement. By combining analytical rigor with visual evidence and contextual reasoning, the report demonstrates how business statistics can be used as a decision-support tool in real life scenarios.

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3. Overview of the Analytical Approach

This report is divided into four independent problem statements, each addressing a different decision-makFor every problem, the analysis begins by understanding the situation and identifying what decision needs to be supported using data.

Based on the nature of the problem, the appropriate statistical tools were applied. These include basic probability calculations, normal distribution analysis, hypothesis testing, and ANOVA. Python was used to perform the calculations and generate charts, which are included throughout the report to support the findings visually.

Rather than focusing only on formulas or test results, the emphasis is on interpretation. Each result is explained in simple terms and linked back to the original business question so that the conclusions are easy to follow and practically meaningful.

The sections that follow present each problem separately, along with the relevant charts, observations, and business implications.

4. Problem 1: Foot Injuries and Player Positions

4.1 Background

A physiotherapist working with a male football team wanted to understand whether foot injuries are related to player positions. Since different positions involve different levels of physical strain and movement patterns, identifying injury-prone roles can help in planning targeted prevention and recovery strategies.

4.2 Data and Method Used

The data consists of counts of injured and non-injured players across four playing positions: striker, forward, attacking midfielder, and winger. Probability concepts were used to analyze the distribution of injuries across positions.

	Striker	Forward	Attacking Midfielder	Winger	Total
Players Injured	45	56	24	20	145
Players Not Injured	32	38	11	9	90
Total	**77**	**94**	**35**	**29**	**235**

4.3 Key Findings

4.3.1 What is the probability that a randomly chosen player would suffer an injury?

Calculation:

$$\text{P(Injured)} = \frac{Number\ of\ Injured\ Players}{Total\ Number\ of\ Players}$$

$$\text{P(Injured)} = \frac{145}{235} = 0.617$$

Observation: This implies that roughly 62% of the team's players have suffered a foot injury. Injuries are therefore quite common and not limited to a small group of players.

4.3.2 What is the probability that a player is a Forward or a Winger?

Total Forwards = 94
Total Wingers = 29

Calculation:

$$\text{P(Forward or Winger)} = \frac{94+29}{235}$$

$$\text{P(Forward or Winger)} = \frac{123}{235} = 0.523$$

Observation: About 52% of the players play either as forwards or wingers. This shows that more than half of the team is involved in attacking positions.

4.3.3 What is the probability that a player is a Striker and Injured?

Injured Strikers = 45

Calculation:

$$P(\text{Striker and Injured}) = \frac{45}{235}$$

$$P(\text{Striker and Injured}) = 0.191$$

Observation: Nearly 19% of all players are injured strikers. This is a noticeable share and suggests that injuries among strikers contribute significantly to the overall injury count.

4.3.4 What is the probability that an injured player is a Striker?

Calculation:

$$P(\text{Striker} \mid \text{Injured}) = \frac{Injured\ Strikers}{Total\ Injured\ Players}$$

$$P(\text{Striker} \mid \text{Injured}) = \frac{45}{145} = 0.310$$

Observation: Among injured players, around 31% are strikers. This indicates that strikers form a large portion of the injured group compared to other positions.

4.4 Overall Summary

- A large proportion of players in the team have suffered foot injuries, showing that injuries are fairly common.
- More than half of the players belong to attacking positions such as forwards and wingers.
- A noticeable share of all players are injured strikers, indicating that injuries among strikers are not rare.
- Among injured players, strikers form a significant portion compared to other positions.
- Overall, injuries appear to be more concentrated in attacking roles, especially strikers, rather than being evenly spread across positions.

5. Problem 2: Breaking Strength of Gunny Bags

5.1 Background

A cement manufacturing company uses gunny bags for packaging cement during transportation and storage. If these bags are too weak, they can tear during handling, leading to material loss, spillage, and financial loss. To avoid this, the company wants to understand how strong the gunny bags are and how likely it is for bags to fall below or above certain strength limits.

The breaking strength of the gunny bags is assumed to follow a normal distribution, which allows the use of probability and standard normal (Z-score) calculations to estimate the proportion of bags in different strength ranges.

5.2 Data Overview

- The breaking strength of gunny bags follows a normal distribution.
- Mean breaking strength (μ) = 5 kg/cm^2
- Standard deviation (σ) = 1.5 kg/cm^2
- The analysis focuses on finding the proportion of bags:
  - Below certain strength levels
  - Above the minimum acceptable strength
  - Within specific ranges
  - Outside a wide acceptable range

These values are used to calculate probabilities using Z-scores and the standard normal distribution table.

5.3 Key Findings

5.3.1 What proportion of bags have a breaking strength less than 3.17 kg/cm^2?

Calculation:

$$Z = \frac{3.17-5}{1.5} = -1.22$$

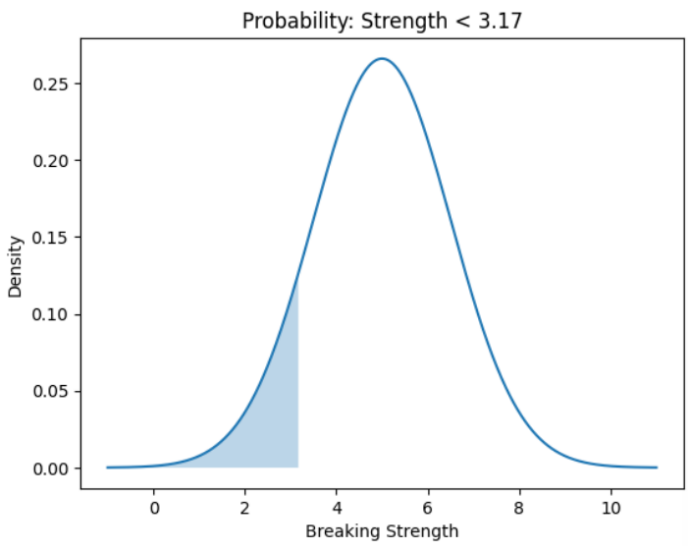
From the standard normal table:

$$P(Z < -1.22) = 0.111$$

Result:
Proportion = 0.111 (11.1%)

Observation:

- Only about 11% of the gunny bags have a breaking strength below 3.17 kg/cm^2, indicating that very weak bags are from a small portion of the total.
- The shaded area on the left side of the curve represents bags with breaking strength below 3.17. This area is relatively small compared to the full curve, indicating that only a small proportion of bags fall within this low-strength range.



5.3.2 What proportion of the gunny bags have a breaking strength of at least 3.6 kg/cm^2?

Calculation:

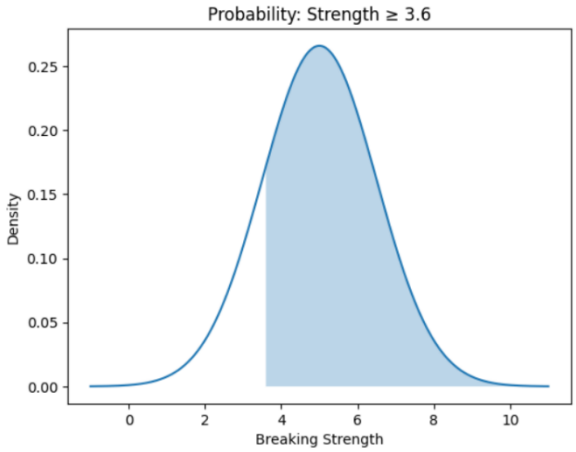
$$Z = \frac{3.6-5}{1.5} = -0.93$$

From the Z-table:

$$P(Z < -0.93) = 0.176$$

$$P(X \geq 3.6) = 1 - 0.176 = 0.824$$

Result:
Proportion = 0.824 (82.4%)



Observation:

- Around 82% of the bags meet or exceed a strength of 3.6 kg/cm^2, showing that most bags are strong enough for handling and transport.
- The shaded area to the right of 3.6 covers most of the curve, showing that a large proportion of gunny bags have a breaking strength of at least 3.6.

5.3.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg/cm^2?

Calculation:

For 5:

$$Z = \frac{5-5}{1.5} = 0$$

For 5.5:

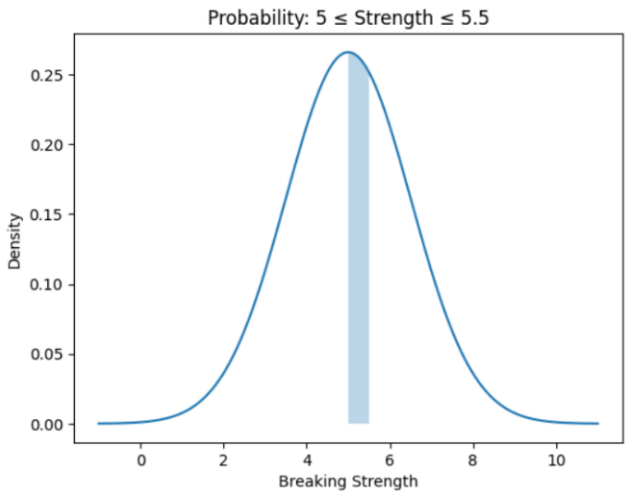
$$Z = \frac{5.5-5}{1.5} = 0.33$$

From the Z-table:

$$P(Z < 0.33) = 0.629$$

$$P(5 \leq X \leq 5.5) = 0.629 - 0.500 = 0.129$$

Result:
Proportion = 0.129 (12.9%)



Observation:

- Only about 13% of the bags fall in this narrow range near the mean, which is expected in a normal distribution where values are spread out.
- The shaded region between 5 and 5.5 is quite narrow, indicating that only a small proportion of gunny bags have breaking strength in this specific range around the mean.

5.3.4 What proportion of the gunny bags do NOT have a breaking strength between 3 and 7.5 kg/cm^2?

Calculation:

For 3:

$$Z = \frac{3-5}{1.5} = -1.33$$

For 7.5:

$$Z = \frac{7.5-5}{1.5} = 1.67$$

From the Z-table:

$$P(Z < 1.67) = 0.953$$

$$P(Z < -1.33) = 0.092$$

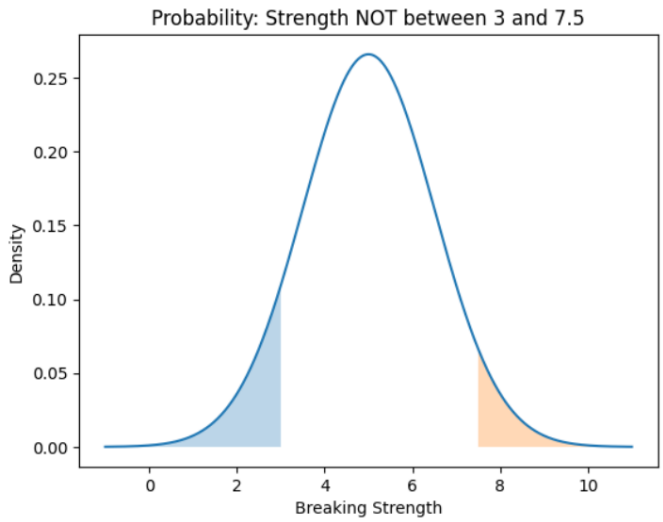
$$P(3 \leq X \leq 7.5) = 0.953 - 0.092 = 0.861$$

$$P(X < 3 \text{ or } X > 7.5) = 1 - 0.861 = 0.139$$

Result:
Proportion = 0.139 (13.9%)

Observation:

- Only about 14% of the bags lie outside this wide range, meaning extreme strength values are relatively uncommon.
- The shaded areas on both ends of the curve represent bags with very low or very high breaking strength. These regions are relatively small, showing that only a small proportion of bags fall outside the range of 3 to 7.5.



5.4 Overall Summary

- Most gunny bags have breaking strength close to the average value, indicating consistent quality.
- Only a small proportion of bags fall below low strength levels, so very weak bags are uncommon.
- A large majority of bags meet or exceed the minimum acceptable strength requirement.
- Bags with extremely low or extremely high breaking strength are rare.
- Overall, the distribution suggests that the current quality of gunny bags is reliable, with issues limited to a small fraction of the production.

6. Problem 3: Zingaro Stone Printing – Hardness Analysis

6.1 Background

Zingaro Stone printing requires stones with a minimum Brinell hardness index of 150 to ensure good print quality. Stones with hardness below this level may lead to poor print quality or surface damage during printing.

Zingaro has received a batch of polished and unpolished stones and wants to evaluate whether the stones meet the required hardness standards and whether polishing has a significant impact on hardness.

All analysis is carried out at a 5% level of significance.

6.2 Data Description

- Unpolished: Represents the Brinell hardness index of stones that have not undergone any polishing or surface treatment.
- Treated and Polished: Represents the Brinell hardness index of stones after being treated and polished for printing purposes.

6.3 Data Overview

6.3.1 Data Shape

The dataset contains 75 observations and 2 variables, corresponding to hardness values for unpolished and treated-and-polished stones.

6.3.2 Data Background and Content

Both columns contain 75 values with no missing data, and all hardness measurements are stored as numeric values, making the dataset ready for analysis.

#	Column	Non – NULL Count	Dtype
0	Unpolished	75 non NULL	float64
1	Treated and Polished	75 non NULL	float64

6.3.3 Initial Data Preview

The initial preview of the dataset shows hardness values for unpolished and treated-and-polished stones recorded side by side. The values vary across observations, suggesting differences in hardness levels between the two stone types.

The first five rows are displayed below:

	Unpolished	Treated and Polished
0	164.481713	133.209393
1	154.307045	138.482771
2	129.861048	159.665201
3	159.096184	145.663528
4	135.256748	136.789227

6.3.4 Statistical summary of the data

	Unpolished	Treated and Polished
count	75.000000	75.000000
mean	134.110527	147.788117
std	33.041804	15.587355
min	48.406838	107.524167
25%	115.329753	138.268300
50%	135.597121	145.721322
75%	158.215098	157.373318
max	200.161313	192.272856

- Unpolished stones have a lower average hardness compared to treated-and-polished stones.
- The hardness values of unpolished stones show greater variation, as indicated by the higher standard deviation and wider range.
- Some unpolished stones have very low hardness values, falling well below the desired threshold.
- Treated-and-polished stones have a higher mean hardness value, falling well below the desired threshold.
- Treated-and-polished stones have a higher mean hardness and a more compact distribution.
- The higher minimum hardness and lower variability in polished stones suggest that the polishing process improves both hardness and consistency.

6.4 Checking consistency of data

- A consistency check was performed on the dataset before analysis.
- All values were found to be complete and correctly formatted.
- Minor inconsistencies were observed in the column name “Unpolished”, where extra spaces were present.

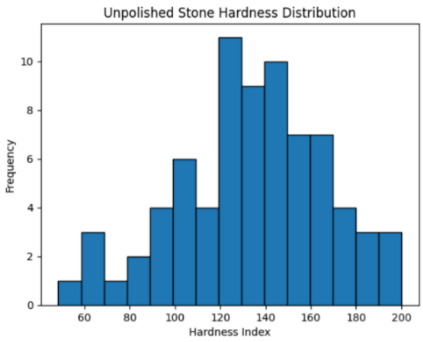
```
Index(['Unpolished ', 'Treated and Polished'], dtype='object')
Index(['Unpolished', 'Treated and Polished'], dtype='object')
```

- These extra spaces were identified and corrected to ensure smooth data handling and accurate analysis.

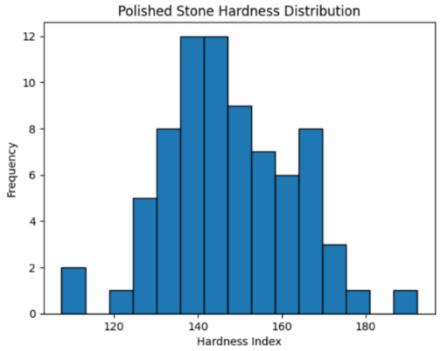
6.5 Visual Representation of Data

Histograms were plotted separately for unpolished and polished stones to understand how hardness values are distributed.

- The distribution of unpolished stone hardness is fairly broad, with values ranging from very low to relatively high. A noticeable number of stones fall below the desired hardness threshold of 150, and the spread of the data indicates considerable variability. This suggests that unpolished stones do not have consistent hardness and that several samples may be unsuitable for printing without further treatment.



- The polished stone hardness values are more concentrated at higher hardness levels, with most observations exceeding the required threshold of 150. Compared to unpolished stones, the spread is narrower, indicating greater consistency after polishing. This distribution suggests that polishing improves both the overall hardness and uniformity of the tones, making them more suitable for printing.



6.6 Are unpolished stones suitable for printing?

Zingaro Stone Printing requires stones to have a minimum Brinell hardness index of 150 for effective and reliable printing. Based on the initial data exploration, unpolished stones showed a wide spread of hardness values, with several observations falling well below the required threshold. This raised concerns regarding their suitability for printing.

6.6.1 Statistical Evidence

A statistical summary of the unpolished stone hardness values indicated that the mean hardness is below 150, with high variability across samples. The histogram further supported this observation by showing a substantial portion of values clustered below the required hardness level.

To formally assess this concern, a one-sample t-test was conducted at a 5% significance level.

6.6.2 Hypotheses

- Null Hypothesis (H_0): The mean hardness of unpolished stones us greater than or equal to 150.
- Alternative Hypothesis (H_1): The mean hardness of unpolished stones is less than 150.

6.6.3 Test Result

The test produced a p-value less than 0.05, indicating that the observed mean hardness is statistically significantly lower than the required threshold.

6.6.4 Conclusion

Since the p-value is below the significance level, the null hypothesis is rejected.

6.6.5 Observation

There is sufficient statistical evidence to conclude that unpolished stones do not consistently meet the minimum hardness requirement for printing. Therefore, Zingaro is justified in believing that unpolished stones may not be suitable for printing without additional treatment.

6.7 Comparison of Mean Hardness between Polished and Unpolished

To determine whether polishing has an effect on stone hardness, the mean hardness values of unpolished and treated-and-polished stones were compared. Initial inspection of the statistical summaries showed that polished stones have a higher average hardness and lower variability compared to unpolished stones. The histograms further supported this, with polished stones clustered around higher hardness values and fewer low-hardness observations.

6.7.1 Hypotheses

- Null Hypothesis (H_0): The mean hardness of polished stones is equal to the mean hardness of unpolished stones.
- Alternative Hypothesis (H_1): The mean hardness of polished stones is different from the mean hardness of unpolished stones.

6.7.2 Test used

An independent two-sample t-test was conducted at a 5% significance level on unpolished stones.

6.7.3 Test Result

The test yielded a p-value less than 0.05, indicating a statistically significant difference between the two mean hardness values.

6.7.4 Conclusion

Since the p-value is below the chosen significance level, the null hypothesis is rejected.

6.7.5 Observation

The results confirm that polished stones have a significantly higher mean hardness than unpolished stones. This indicates that the polishing process has a measurable impact on stone hardness and improves suitability for printing.

6.8 Overall Summary

- The hardness values of unpolished stones show considerable variation, with several observations falling below the required hardness threshold of 150.
- Statistical analysis confirms that the average hardness of unpolished stones is below the minimum level needed for printing.
- Polished stones exhibit higher and more consistent hardness values compared to unpolished stones.
- The differences in mean hardness between polished and unpolished stones are statistically significant.
- The polishing process improves both the hardness and uniformity of stones.
- Overall, polished stones are more suitable for printing, while unpolished stones may require additional treatment before use.

7. Problem 4: Dental Implant Hardness Analysis

7.1 Background

In dental implant procedures, the hardness of metal implants is an important quality characteristic, as it affects durability, performance, and long-term reliability. Implant hardness can be influenced by multiple factors, including the method used during implantation, the temperature at which the metal is treated, the type of alloy used, and the dentist performing the procedure. Dentists may prefer certain methods and may also perform better with techniques they are more familiar with, which can further impact the final hardness of the implant.

Given these multiple influencing factors, it is necessary to understand how implant hardness varies across dentists and methods, both individually and in combination. This analysis focuses on identifying whether these factors have a statistically significant effect on implant hardness and whether any interaction exists between them.

7.2 Data Description

The dataset contains experimental data related to the hardness of dental implants:

- Dentist: Identifies the dentist who performed the implant procedure.
- Method: Represents the implant method used during the procedure.
- Alloy: Indicates the type of alloy used in the implant. Two different alloy types are included in the dataset, and analyses are conducted separately for each alloy.
- Temp: Denotes the temperature at which the implant material was treated.
- Response: Represents the hardness of the dental implant. This is the response variable of interest and is measured on a numerical scale.

7.3 Data Overview

7.3.1 Data Shape

The dataset contains 90 observations and 5 variables, covering information on dentists, methods, alloys, treatment temperature, and implant hardness.

7.3.2 Data Background and Content

All five variables contain complete data with no missing values, and each variable is stored in a consistent numeric format. This confirms that the dataset is clean and suitable for further statistical analysis.

#	Column	Non-NULL Count	Dtype
0	Dentist	90 non-NULL	int64
1	Method	90 non-NULL	int64
2	Alloy	90 non-NULL	int64
3	Temp	90 non-NULL	int64
4	Response	90 non-NULL	int64

7.3.3 Initial Data Preview

The first few records show implant hardness values along with the corresponding dentist, method, alloy,type, and treatment temperature. Each row represents one implant observation, giving a clear view of how the response variable is recorded against multiple influencing factors.

	Dentist	Method	Alloy	Temp	Response
0	1	1	1	1500	813
1	1	1	1	1600	792
2	1	1	1	1700	792
3	1	1	2	1500	907
4	1	1	2	1600	792

7.3.4 Statistical Summary of the Data

- Implant hardness values show a wide range, indicating noticeable variation across observations.
- The response variable includes both relatively low and very high hardness values, suggesting differences across experimental conditions.
- Dentists, methods, and alloy types are evenly represented in the dataset, as reflected by similar counts.
- Treatment temperatures are concentrated around specific levels, indicating controlled processing conditions.
- The variability in hardness suggests that factors such as method and dentist may influence the final implant hardness.

7.4 Segregation of Alloys

- The dataset contains two alloy types, with an equal number of observations for each.
- Alloy 1 and Alloy 2 each have 45 records.
- The balanced distribution across alloys allows for a fair comparison between the two alloy types.
- Separate analysis by alloy is appropriate and supported by the data structure.

```
Alloy
1    45
2    45
Name: count, dtype: int64
```

7.5 How does the hardness of implants vary depending on the dentist?

To examine whether implant hardness varies across dentists, the analysis was carried out separately for each alloy. Initial descriptive statistics showed some variation in mean hardness values across dentists. Normality and homogeneity of variance checks were performed; even where assumptions were borderline, the analysis was continued as instructed.

7.5.1 Hypotheses

- Null Hypothesis (H_0): Mean implant hardness is the same for all dentists.
- Alternative Hypothesis (H_1): At least one dentist has a different mean implant hardness.

7.5.2 Test Used

One-way ANOVA(Dentist as factor)

7.5.3 Results and Interpretation

- Alloy 1: The ANOVA result showed no statistically significant difference in hardness across dentists. Post-hoc Tukey tests also indicated no significant pairwise differences.
- Alloy 2: Similar results were observed, with no statistically significant differences in hardness between dentists.

7.5.4 Observation

For both alloys, implant hardness does not vary significantly based on the dentist alone. This suggests that dentist-specific preferences or techniques do not independently influence hardness outcomes.

7.6 How does the hardness of implants vary depending on the methods?

The effect of the implant method on hardness was examined separately for each alloy. Mean hardness values differed visibly across methods, which was further assessed using statistical testing.

7.6.1 Hypotheses

- Null Hypothesis (H_0): Mean implant hardness is the same across all methods.
- Alternative Hypothesis (H_1): At least one method produces a different mean hardness.

7.6.2 Test Used

One-way ANOVA (Method as factor)

7.6.3 Results and Interpretation

- Alloy 1: The ANOVA test showed a statistically significant difference in hardness across methods. This indicates that the method used has a measurable impact on implant hardness.
- Alloy 2: A statistically significant difference was also observed, with the method having an even stronger effect on hardness.

7.6.4 Observation

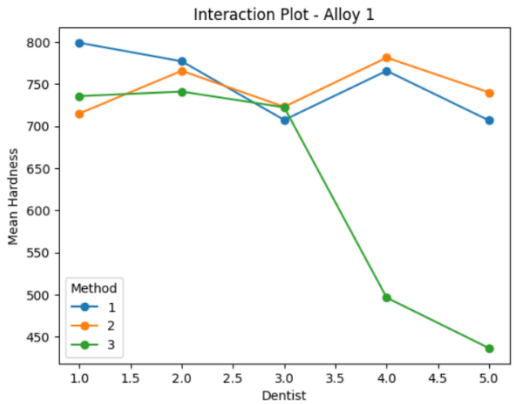
For both alloys, the implant method plays a significant role in determining hardness. This suggests that process-related factors linked to the method are important drivers of implant quality.

7.7 What is the interaction effect between the dentist and method for each alloy?

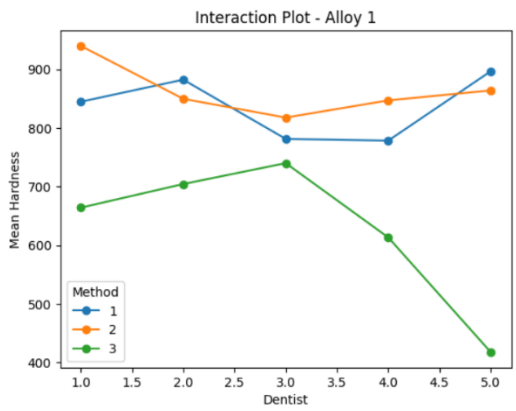
To study whether the effect of the method varies across dentists, interaction plots were created separately for each alloy. These plots visually compare mean hardness across methods for each dentist.

7.7.1 Visual Analysis

- Alloy 1:
  - The lines for different methods are not parallel, indicating an interaction between the dentist and method for Alloy 1.
  - For Methods 1 and 2, mean hardness values vary moderately across dentists.
  - Method 3 shows a sharp drop in hardness for Dentists 4 and 5 compared to earlier dentists.
  - This suggests that the effect of the method on hardness depends on the dentist performing the procedure.
  - Overall, Alloy1 shows a noticeable interaction between the dentist and the method.



- Alloy 2:
  - The lines for the three methods are clearly non-parallel, indicating an interaction between the dentist and method.
  - Methods 1 and 2 maintain relatively higher hardness values across most dentists.
  - Method 3 shows a steady increase up to Dentist 3, followed by a sharp decline for Dentists 4 and 5.
  - The relative performance of methods changes across dentists, rather than remaining consistent.
  - This pattern suggests that the effect of the method on implant hardness depends on the dentist performing the procedure.



7.7.2 Observation

For Alloy 1, there is a noticeable interaction effect, meaning that certain dentists achieve better hardness outcomes with specific methods. For Alloy 2, the interaction effect appears limited, indicating more consistent performance across dentists.

7.8 How does the hardness of implants vary depending on dentists and methods together?

A two-way ANOVA was conducted for each alloy to jointly assess the effects of dentist, method, and their interaction on implant hardness.

7.8.1 Hypotheses

- Null Hypothesis (H_0): Dentist, method, and their interaction do not affect implant hardness.
- Alternative Hypothesis (H_1): At least one of these factors has a significant effect.

7.8.2 Test Used

A two-way ANOVA with an interaction term

7.8.3 Results and interpretation

- Alloy 1: Dentist, method, and the interaction between them were found to be statistically significant. This indicates that hardness depends not only on individual factors but also on how the dentist and the method work together.
- Alloy 2: The method was statistically significant, while the dentist and interaction effects were not. This suggests that hardness is primarily driven by the method used rather than who performs it.

7.8.4 Observation

The combined analysis shows that Alloy 1 is more sensitive to both human and process-related factors, while Alloy 2 behaves more consistently and is mainly influenced by method choice.

7.9 Overall Summary

- Implant hardness shows noticeable variation across the dataset, indicating that multiple factors influence the final outcome.
- The dentist alone does not lead to consistent differences in implant hardness for either alloy.
- The method used had a clear impact on implant hardness, making it an important factor in determining implant quality.
- For Alloy 1, the interaction between the dentist and method is evident, meaning the effectiveness of a method varies depending on the dentist.
- For Alloy 2, the interaction effect is weaker, and hardness is mainly influenced by the method rather than the dentist.

- Overall, Alloy 1 appears more sensitive to both human and process-related factors, while Alloy 2 shows more stable behaviour across dentists.

8. Conclusion

In this project, I used statistical methods to analyze different real-world problems and support decision-making using data. Each problem addressed a different scenario, but together they showed how probability, distributions, hypothesis testing, and ANOVA can be applied to understand variation and uncertainty.

In the football injury analysis, I observed that injuries are quite common and are more concentrated in attacking positions, especially among strikers. This indicates that injuries are not evenly distributed across positions and that certain roles may require more focused injury prevention and management.

The analysis of gunny bag breaking strength showed that most bags meet acceptable strength levels, with only a small proportion falling into weak or extreme categories. The probability calculations and visual distributions helped me understand where potential quality risks exist, while also showing that the overall packaging quality is fairly consistent.

For the stone hardness analysis at Zingaro Company, I found that unpolished stones often do not meet the minimum hardness requirement for printing. Polished stones, however, showed higher and more consistent hardness values. The statistical tests confirmed that polishing has a clear impact on stone hardness and is necessary for reliable printing quality.

In the dental implant analysis, I found that implant hardness is influenced more by the method used than by the dentist alone. For one alloy, the interaction between dentist and method played an important role, while the other alloy showed more stable behavior driven mainly by process conditions. This highlighted the importance of selecting appropriate methods and understanding how different factors interact.

Overall, this project helped me understand how statistical analysis can be used to move beyond assumptions and provide clear, data-driven insights that support better decisions across different contexts.